

Chapter 1 Algebra 1

$2 \times 2 \times 2 = 2^3$ ← Index ← Base		$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$ $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$ $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$	Index Laws $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{m \times n}$ $a^0 = 1$ $a^{-m} = \frac{1}{a^m}$
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Chapter 2 Linear Equations

$x + 3 > 1$ $x > 1 - 3$ $x > -4$		Expressions such as $H = 86d + 65$ are linear because the highest power of d is 1 ($d = d^1$).
If you multiply or divide both sides of an inequation by a negative? $7 > 3$ $7 \times 2 \quad 3 \times 2 \quad < \text{becomes} >$ $-14 < -6 \quad > \text{becomes} <$	Substitute either x or y from one equation into the other equation $x + y = 53 \quad \dots (1)$ $x = 2y + 5 \quad \dots (2)$	Eliminate either x or y by adding or subtracting the equations. $3x + 3y = 51 \quad \dots (1)$ $x - 3y = 5 \quad \dots (2)$

Chapter 3 Area & Volume

Area formulas Area rectangle = length $\times$ breadth Area triangle = $\frac{1}{2}$ base $\times$ height Area circle = $\pi \times \text{radius}^2$	Prisms are three-dimensional shapes that have a constant cross-section. $V_{\text{prism}} = \text{Area of base} \times \text{height}$
Right Pyramid 10A $V_{\text{pyramid}} = \frac{1}{3} \times \text{Area base} \times \text{height}$ 	Sphere 10A $V_{\text{sphere}} = \frac{4}{3} \pi r^3$

Chapter 4 Indices & Logs 10A

Surd Rule $\sqrt{x} \times \sqrt{y} = \sqrt{xy}$	$(8x^3)^{\frac{2}{3}} = \frac{1}{(\sqrt[3]{8x^3})^2}$ $= \frac{1}{(2x)^2}$ $= \frac{1}{4x^2}$	$\log_b x + \log_b y = \log_b xy$
Surd Rule $\sqrt{xy} = \sqrt{x} \times \sqrt{y}$	$\log_{10} 100 = 2$ $100 = 10^2$	$\log_b x^d = d \log_b x$ $\log_b x - \log_b y = \log_b \frac{x}{y}$